

Implementation Roadmap

Project: HisabDo Website Enhancement

Department: AI/ML

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Role: AI/ML Team Lead & AI Architect

1. Purpose

This roadmap defines the planned implementation path for **HisabDo AI Copilot** after the current research, architecture, documentation, and POC stages.

The objective is to move from:

Research & Documentation



Improved AI POC



Basic AI Implementation



RAG Integration



Multilingual AI



Financial Intelligence



AI Engagement



Advanced AI Capabilities

The roadmap follows a phased approach to avoid implementing all AI features simultaneously. Each phase should be validated before moving to the next stage.

2. Current Stage Review

Before Day 26+ implementation begins, the team should review completed work.

The review should include:

- AI Copilot architecture
- Current AI workflow
- AI Chatbot concept
- FAQ Assistant research
- AI-powered Help and Support
- AI content recommendations
- User engagement ideas
- AI feasibility research
- AI Enhancement Proposal
- Improved AI POC
- AI use cases
- Technical documentation
- Integration architecture
- Knowledge base requirements

The purpose is to identify:

Completed

In Progress

Technically Ready

Requires Research

Blocked

Future Scope

3. Implementation Priority

The recommended priority is:

1. Basic AI Assistant



2. FAQ and Knowledge Support



3. RAG Integration

- ↓
4. Multilingual Support
- ↓
5. Financial Intelligence
- ↓
6. AI Engagement
- ↓
7. Advanced AI

The first priority should be a stable and controlled AI foundation.

4. Phase 1 — Basic AI Assistant

Objective

Create a basic working AI assistant for HisabDo.

Main Components

- Chat UI
- Basic AI API
- Basic backend communication
- Intent detection
- FAQ support
- Controlled responses
- Basic testing

Proposed Flow

User



Chat UI



Backend API



AI Service



Intent Detection



FAQ / Response Logic



User Response

Key Milestones

- Chat interface ready
- Basic API communication working
- Sample user queries tested
- Basic intent detection available
- FAQ responses working
- Error handling tested

Exit Criteria

Phase 1 should be considered complete when the assistant can handle basic HisabDo-related questions reliably.

5. Phase 2 — RAG Integration

Objective

Enable the AI assistant to answer questions using verified HisabDo knowledge.

Main Components

- Knowledge Base
- Document collection
- Content verification
- Document cleaning
- Text chunking
- Embeddings
- FAISS or vector search
- Similarity search

- RAG response generation

Proposed Flow

HisabDo Content



Cleaning



Chunking



Embeddings



Vector Store



User Question



Similarity Search



Relevant Context



AI Response

Dependencies

- Verified website content
- FAQs
- Product documentation
- Feature descriptions
- Help and support content

Testing

The team should test:

- Relevant information retrieval
- Incorrect query handling

- Missing information
- Unsupported questions
- Response quality

Exit Criteria

The AI should retrieve relevant HisabDo knowledge before generating responses.

6. Phase 3 — Multilingual Support

Objective

Enable users to interact with the AI in their preferred language.

Initial Languages

- English
- Roman Urdu
- Urdu

Main Components

- Language detection
- Input normalization
- Intent understanding
- Same-language response generation

Flow

User Input



Language Detection



Input Normalization



Intent Detection



AI Processing



Response

Special Focus

Roman Urdu should be tested for different spelling variations.

Example:

udhar

udhaar

udharr

kitna

kitnay

kitny

Testing Milestone

The team should prepare multilingual test queries and evaluate response consistency.

7. Phase 4 — Financial Intelligence

Objective

Enable controlled AI assistance for financial questions.

Features

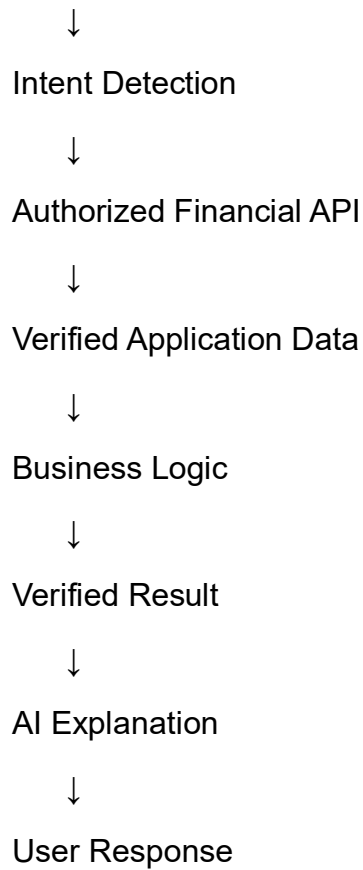
- Financial intent detection
- Verified API tools
- Customer balance queries
- Expense queries
- Receivable analysis
- Financial explanations

Security Principle

AI does not directly access the database.

The architecture should follow:

User Question



Testing

The team should test:

- User authorization
- Correct financial data
- Invalid customer names
- Missing information
- API failures
- Unauthorized access

Exit Criteria

Financial responses must be based on verified system data.

8. Phase 5 — AI Engagement

Objective

Use AI to improve user engagement and product usage.

Features

- Smart suggestions
- Personalized recommendations
- Dashboard insights
- Feature discovery
- Follow-up suggestions
- Context-aware assistance

Examples

Pending balance detected



Suggest customer follow-up

Monthly expenses increased



Show dashboard insight

User has not used a feature



Suggest relevant feature

Priority

Initial engagement features should use simple and controlled logic.

More advanced personalization can be introduced later.

9. Phase 6 — Advanced AI

Objective

Introduce advanced AI capabilities after the core AI system is stable.

Future Features

- Voice AI
- Predictive insights
- Advanced recommendations
- Personalized AI
- Advanced conversation memory

- Local AI optimization
- Offline AI support

Important Requirement

These features should not begin until:

- Core AI is stable
 - RAG quality is acceptable
 - Security is tested
 - Financial access is controlled
 - User feedback is available
-

10. Module Dependencies

The modules should follow this dependency order:

Basic Architecture



AI Assistant



Knowledge Base



RAG



Multilingual Support



Financial APIs



Financial Intelligence



AI Engagement



Advanced AI

For example, RAG should not be treated as fully complete without a prepared and verified knowledge base.

11. Development Milestones

Milestone 1

Basic AI Foundation

- Basic assistant
- API connection
- Intent handling

Milestone 2

Knowledge Intelligence

- Knowledge base
- FAQ data
- RAG prototype

Milestone 3

Multilingual Intelligence

- English
- Roman Urdu
- Urdu

Milestone 4

Financial Intelligence

- Financial intent detection
- Verified API tools
- Controlled responses

Milestone 5

AI Engagement

- Suggestions
- Recommendations
- Dashboard insights

Milestone 6

Advanced AI

- Voice
 - Predictive capabilities
 - Offline optimization
-

12. Testing Milestones

Every phase should include testing.

Functional Testing

- Expected queries
- Unexpected queries
- Invalid input
- Unknown questions

AI Response Testing

- Response relevance
- Response consistency
- Unsupported answers
- Fallback behavior

Multilingual Testing

- English
- Roman Urdu
- Urdu
- Mixed-language input

Integration Testing

- Frontend
- Backend
- AI service
- API communication

Security Testing

- Authentication
 - Authorization
 - Data isolation
 - Financial data protection
-

13. Integration Milestones

The proposed integration sequence is:

AI Service



Backend Integration



Frontend Chat Integration



Knowledge Base Integration



RAG Integration



Financial API Integration



Dashboard AI Integration

Integration should be incremental rather than attempting complete system integration at once.

14. Module Ownership

Module ownership should be clearly assigned before development begins.

Suggested structure:

AI Lead / Architect

- Architecture decisions
- Integration planning

- Technical review
- Security review
- Roadmap management

Chatbot Engineer

- Chatbot service
- AI API
- Conversation handling
- Response generation

RAG Engineer

- Knowledge base
- Document processing
- Embeddings
- Vector search
- RAG evaluation

AI Research / Data Engineer

- AI datasets
- FAQ collection
- Multilingual testing
- AI evaluation
- Research and documentation

Ownership can be adjusted according to team availability and future requirements.

15. Documentation Requirements

Every implemented module should include:

- Module overview
- Architecture
- Setup requirements
- API documentation
- Input/output format

- Dependencies
- Test cases
- Known limitations
- Future improvements

Recommended documentation:

README

Architecture

API Documentation

Setup Guide

Testing Guide

Known Issues

Roadmap

16. GitHub Contribution Expectations

Each implementation task should include meaningful GitHub activity.

Expected practices:

- Clear commit messages
- Separate branches where required
- Regular commits
- Pull request reviews
- Updated documentation
- No unnecessary duplicate work

Suggested structure:

feature/ai-chatbot

feature/rag

feature/multilingual

feature/financial-ai

feature/ai-engagement

GitHub activity should demonstrate actual module contribution rather than only one final bulk upload.

17. Technical Blockers and Risks

Potential blockers include:

- Incomplete website content
- Missing FAQs
- Limited backend APIs
- Unclear financial data structure
- Local hardware limitations
- LLM performance limitations
- Multilingual accuracy issues
- Roman Urdu spelling variation
- Integration complexity
- Financial data security risks

Each blocker should be documented with:

Issue
Impact
Priority
Owner
Possible Solution
Status

18. Day 26+ High-Level Plan

Phase 1

Basic AI Assistant

+

Basic Chat UI

+

AI API

+

Intent Detection

+

FAQ Support

+

Testing

Phase 2

RAG Integration

+

Knowledge Base

+

Document Processing

+

Embeddings

+

FAISS

+

Similarity Search

Phase 3

Multilingual Support

+

English

+

Roman Urdu

+

Urdu

+

Input Normalization

Phase 4

Financial Intelligence

+

Financial Intent

+

Verified APIs

+

Balance Queries

+

Expense Analysis

Phase 5

AI Engagement

+

Suggestions

+

Recommendations

+

Dashboard Insights

+

Feature Discovery

Phase 6

Advanced AI

+

Voice AI

+

Predictive Insights

+

Advanced Personalization

+

19. Final Roadmap Summary

Current Research and POC



Phase 1

Basic AI Assistant



Phase 2

RAG Integration



Phase 3

Multilingual Support



Phase 4

Financial Intelligence



Phase 5

AI Engagement



Phase 6

Advanced AI



Complete HisabDo AI Copilot

20. Conclusion

The Day 26+ implementation roadmap provides a structured path from the current **research and POC stage** toward the future **HisabDo AI Copilot**.

The recommended approach is to first build a stable AI foundation, then add knowledge intelligence, multilingual capabilities, verified financial intelligence, engagement features, and finally advanced AI functionality.

The roadmap should remain flexible. Features should only move to the next development phase when their dependencies, testing requirements, technical readiness, and security requirements are satisfied.

Current work is planning and roadmap documentation. Actual coding and implementation can begin in the respective development phases.